INSTITUTE OF DISTANCE LEARNING

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI, GHANA

ME 361
DYNAMICS OF MACHINERY

Lecture 2

RIGID BODY GEOMETRY

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UNIT 2: RIGID BODY GEOMETRY

Unit content

- **Session 2-1: Centre of Mass**
  - 1-1.1 Summation Method
  - 1-2.2 Integration Method
  - 1-2.3 Velocity and Acceleration of Centre of Mass
- **Session 2-2: Mass Moments and Products of Inertia**
  - 2-2.1 Moments of Inertia
  - 2-2.2 Products of Inertia
  - 2-2.3 Inertia Matrix
- **Session 2-3: Transformation of Mass Moments of Inertia**
  - 2-2.1 Moments of Inertia
  - 2-2.2 Products of Inertia
  - 2-2.3 Inertia Matrix

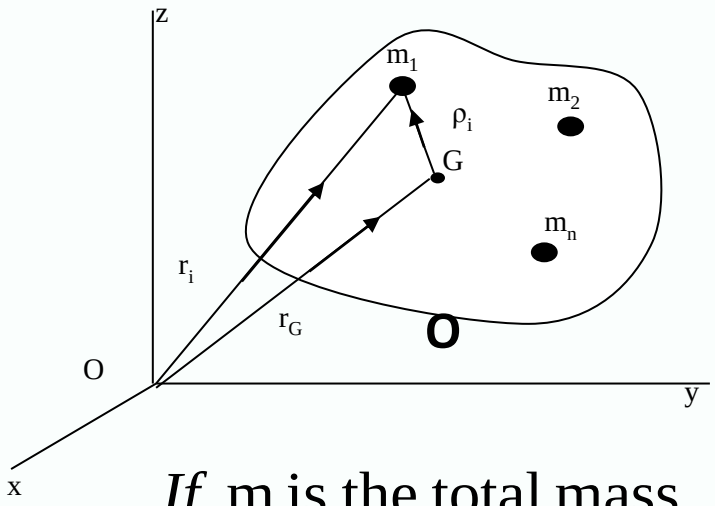
Unit Learning Objectives

After reading this unit you should be able to:

- Determine centre of mass of a rigid using integration and summation methods.
- Determine moment and product of inertia of a body.
- Find inertia matrix of composite body using parallel axis theorem.
- Find inertia matrix of a rigid body after a set of rotation.
- Know when to use integration, parallel axis and rotation methods.
- Find moment of inertia about any arbitrary axis.

Centre of Mass: Summation Method

Consider a rigid consisting of N particles. Let m_i and r_i be the mass and its displacement from the origin O, and i be *ith* particle (i = 1, 2... n). The location of the centre of mass G is defined as



If m is the total mass

$$m_1 + m_2 + m_3 + ... + m_n = m = \sum_{i=1}^{i=n} m_i$$

Taking moment about O

$$m_1 r_1 g + m_2 r_2 g + m_3 r_3 g + ... + m_n r_n g = m r_G g$$

$$r_G = \frac{1}{m}\sum_{i=1}^{n} m_i r_i .....(1)$$

For a body with a uniform density

$$m = \rho V; m_i = \rho V_i \qquad \therefore r_G = \frac{1}{V}\sum_{i=1}^{n} V_i r_i$$

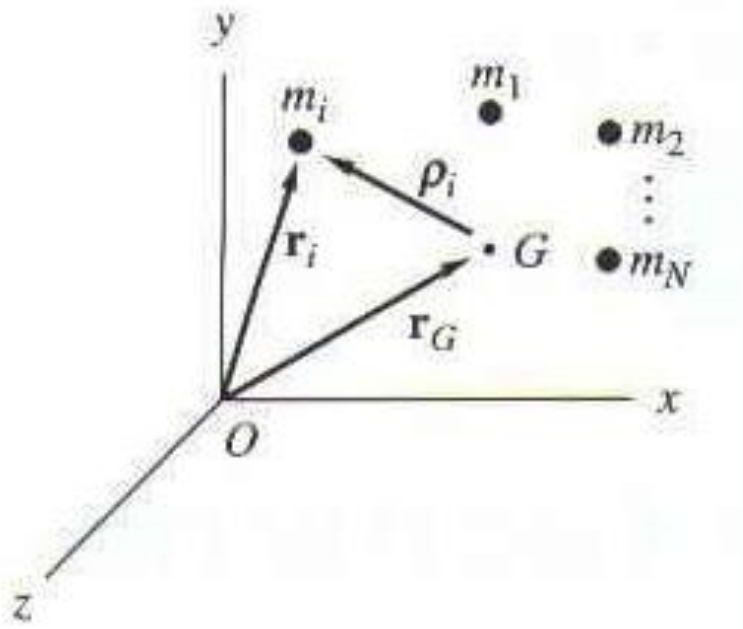
Also for a body with a uniform thickness or with no thickness

$$V = At; V_i = tA_i$$

$$r_G = \frac{1}{A}\sum_{i=1}^{n} A_i r_i$$

Centre of Mass: Summation Method cont'd…

- Note that r_G is a three-dimensional vector representing the x, y and z coordinates of the centre of mass. That is,



$$r_G = x_G i + y_G j + z_G k$$

From the figure $r_i = r_G + \rho_i$

$$r_G = \frac{1}{m}\sum_{i=1}^{n} m_i (r_G + \rho_i) \ldots i \qquad r_G = \frac{1}{m}\sum_{i=1}^{n} m_i r_G + \frac{1}{m}\sum_{i=1}^{n} m_i \rho_i \ldots ii$$

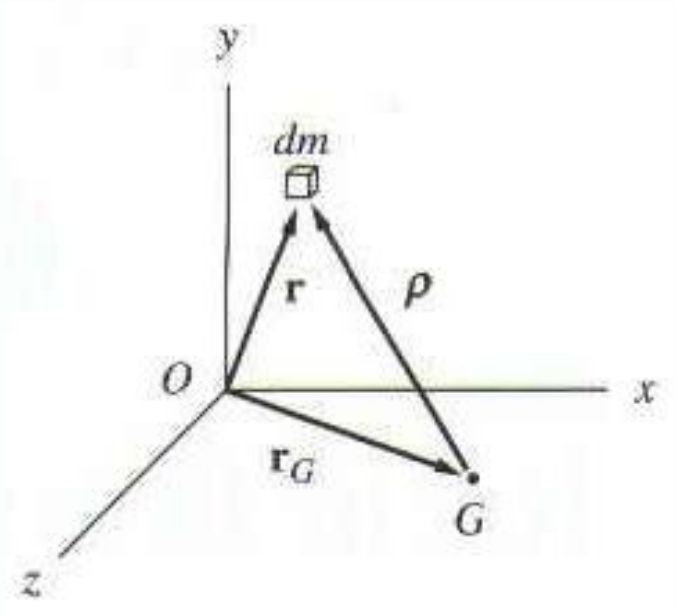
From (1) $r_G = \frac{1}{m}\sum_{i=1}^{n} m_i r_i$

Which implies that

$$\frac{1}{m}\sum_{i=1}^{n} m_i \rho_i = 0$$

Centre of Mass: Integration Method

Consider a infinitesimal element of rigid having a elemental mass dm and located at r from the origin. consisting of N particles. The location of the centre of mass G is defined as



$$r_G = \frac{1}{m}\int_{body} r\,dm \ldots 1$$

where $$m = \int_{body} dm$$

Similarly, r_G is a three-dimensional vector representing the x, y and z coordinates of the centre of mass. That is,

$$r_G = x_G i + y_G i + z_G i$$

From the figure $$r_i = r_G + \rho_i$$

$$r_G = \frac{1}{m}\int_{body}(r_G + \rho)dm \qquad r_G = \frac{1}{m}\int_{body} r_G\,dm + \frac{1}{m}\int_{body}\rho\,dm \ldots 2$$

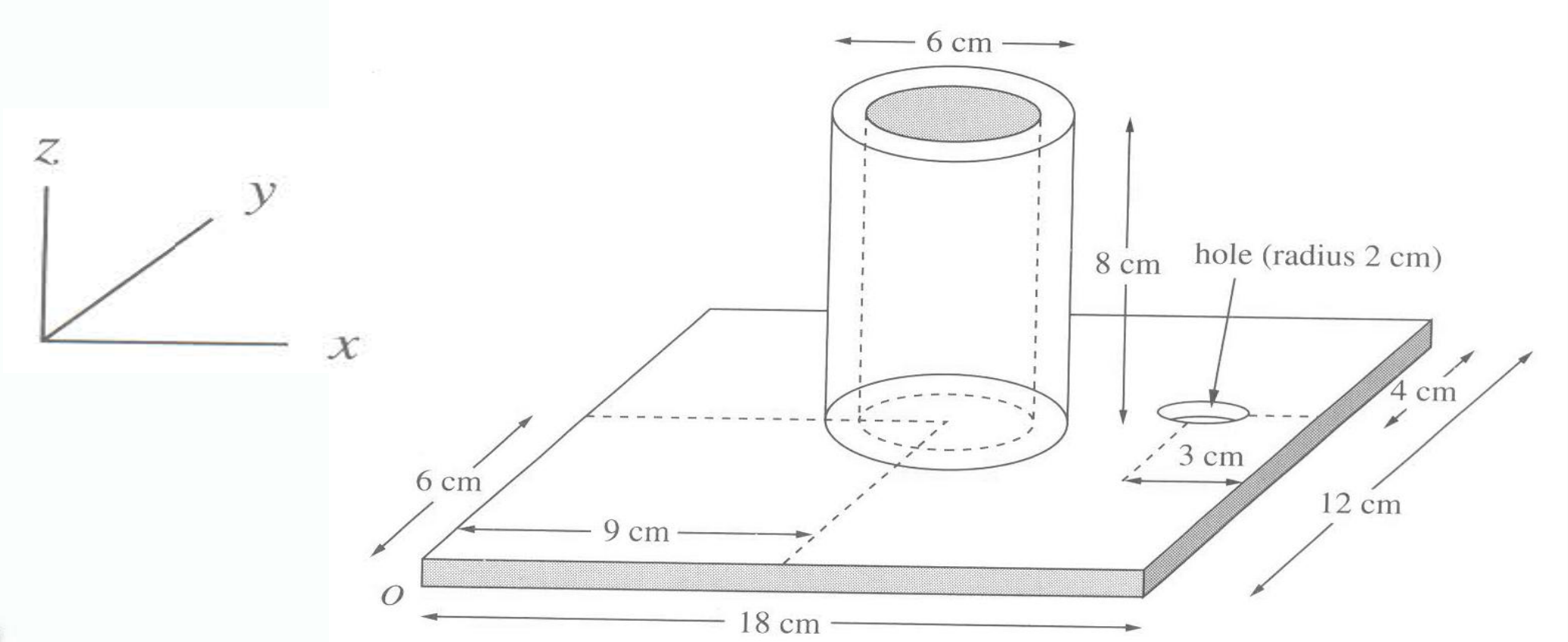
From (1) $$r_G = \frac{1}{m}\int_{body} r\,dm$$

Which implies that

$$\frac{1}{m}\int_{body}\rho\,dm = 0$$

Example 2-1

The figure below shows a decorative pencil holder, made of uniform material of thickness 0.4 cm. Find the center of mass.



Solution

Part	V_i/cm^3	x_i/cm	y_i/cm	z_i/cm	$V_i x_i/cm$	$V_i y_i/cm$	$V_i z_i/cm$
Base(1)	*86.4*	*9*	*6*	*0.2*	*777.6*	*518.4*	*17.28*
Cylinder(2)	*56.3*	*9*	*6*	*4.4*	*506.7*	*337.8*	*247.72*
Hole(3)	*-5.027*	*15*	*8*	*0.2*	*-75.405*	*-40.216*	*-1.0054*
Summation	*137.7*				*1208.895*	*815.984*	*263.995*

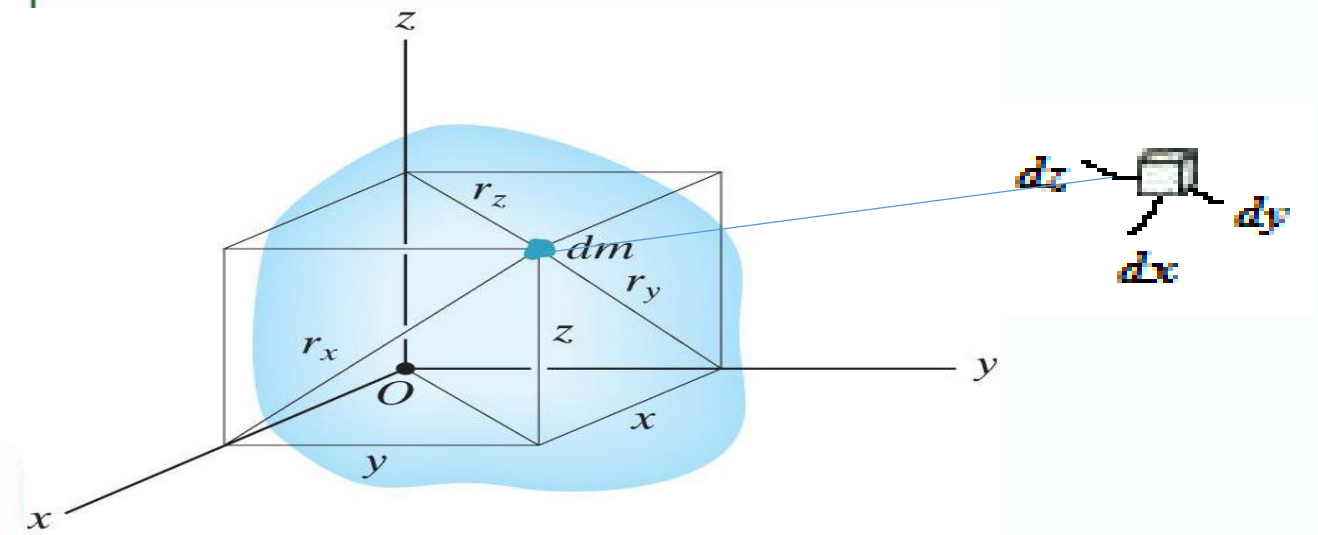
$$V_1 = 12 \times 18 \times 0.4 = 86.4\ cm^3$$
$$V_2 = 8 \times \pi \times (3^2 - 2.6^2) = 56.3\ cm^3$$
$$V_3 = 0.4 \times \pi \times (2^2) = 5.027\ cm^3$$

$$r_G = \frac{\sum_{i=1}^{i=3} V_i r_i}{V} = \frac{1208.895\boldsymbol{i} + 815.984\boldsymbol{j} + 263.9946\boldsymbol{k}}{137.7} = 8.779\boldsymbol{i} + 5.926\boldsymbol{j} + 1.917\boldsymbol{k}\ cm$$

Mass Moments of Inertia

The moment of inertia of a body is a measure of how the mass of the body is distributed about a point in the body, it also describes the resistance of a body to a rotational motion. The moment of inertia of differential element of mass dm about the x axis is

$$dI_{xx} = r_x{}^2 dm = \left(y^2 + z^2\right)dm$$



The moments of inertia about the x, y and z axes may be written as

$$I_{xx} = \int_m r_x{}^2 dm = \int_m \left(y^2 + z^2\right)dm$$

$$I_{yy} = \int_m r_y{}^2 dm = \int_m \left(x^2 + z^2\right)dm$$

$$I_{zz} = \int_m r_{zz}{}^2 dm = \int_m \left(x^2 + y^2\right)dm$$

The moment of inertia is always positive since it is the sum of product of mass and square of distances, which are always positive.

Mass Moments of Inertia

Using rectangular differential element (3-D)

$$dm = \rho dV = \rho\, dx\, dy\, dz$$

Gives

$$I_{xx} = \rho \iiint (y^2 + z^2) dx\, dy\, dz$$

$$I_{yy} = \rho \iiint (x^2 + z^2) dx\, dy\, dz$$

$$I_{zz} = \rho \iiint (x^2 + y^2) dx\, dy\, dz$$

Mass Moments of Inertia

For mass distribution in 2-D

$$dm = \rho dA = \rho\, dx\, dy$$

$$I_{xx} = \rho \iint (y^2 + z^2) dy dz$$

$$I_{yy} = \rho \iint (x^2 + z^2) dx dz$$

$$I_{zz} = \rho \iint (y^2 + x^2) dy dx$$

Mass Moments of Inertia

For mass distribution in 1-D

$$dm = \rho dl$$

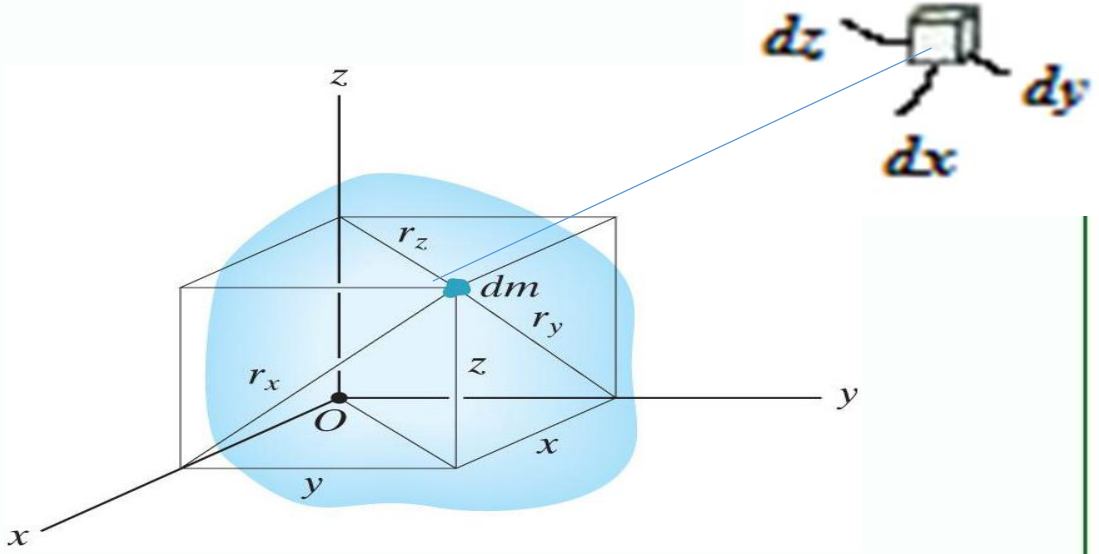
$$I_{xx} = \rho \int (y^2 + z^2)\, dl$$

$$I_{yy} = \rho \int (x^2 + z^2)\, dl$$

$$I_{zz} = \rho \int (y^2 + x^2)\, dl$$

Products of Inertia

The product of inertia for a differential element with respect a set of two perpendicular planes is defined as the product of the mass of the element and shortest perpendicular distances from the planes to the element. In the Figure, the product of inertia for differential element of mass dm with respect to the planes y-z and x-z is



$$dI_{xy} = xydm$$

The product of inertia about the xy, yz and zx planes may be written as

$$I_{xy} = \int_m xydm$$

$$I_{yz} = \int_m yzdm \qquad I_{xz} = \int_m xzdm$$

Product of Inertia cont'd…

Again using rectangular differential element(3-D)

$$dm = \rho dV = \rho\, dx\, dy\, dz$$



Gives

$$I_{xy} = \rho \iiint (xy) dx\, dy\, dz$$

$$I_{yz} = \rho \iiint (yz) dx\, dy\, dz$$

$$I_{xz} = \rho \iiint (xz) dx\, dy\, dz$$

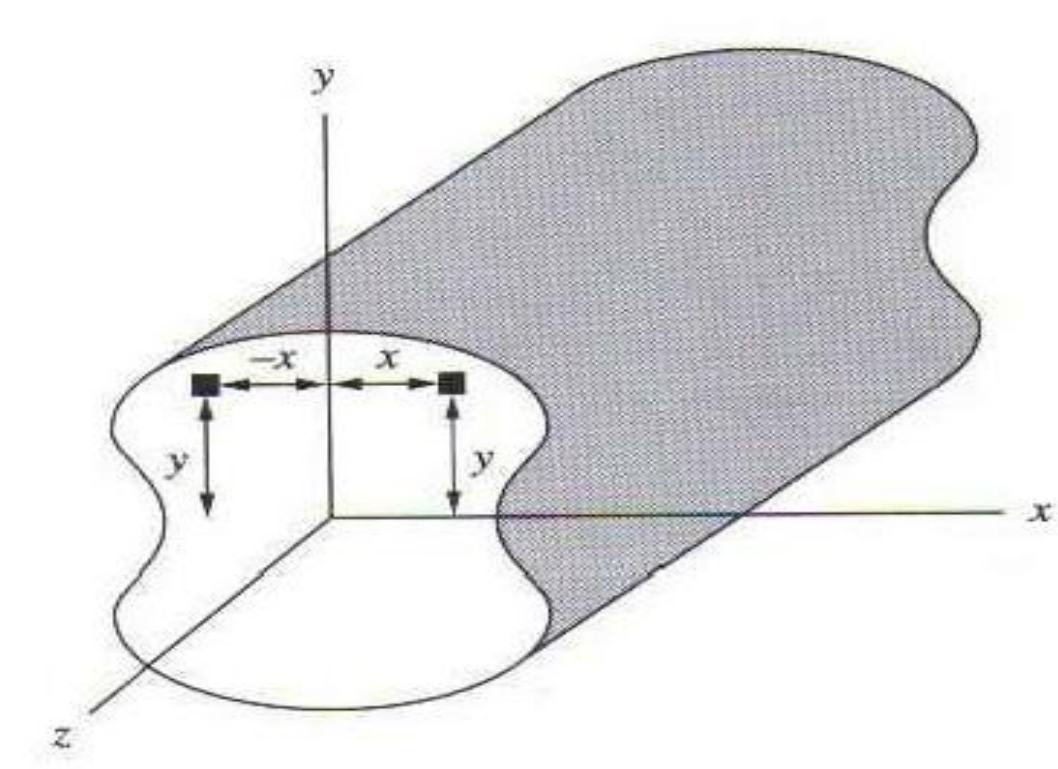
Products of Inertia cont'd…

The product of inertia is an indication of how the mass of a body is distributed about the planes in the body

If any two coordinate axis form a plane of symmetry for a body, then the product of inertia associated with coordinates normal to that planes varnish.

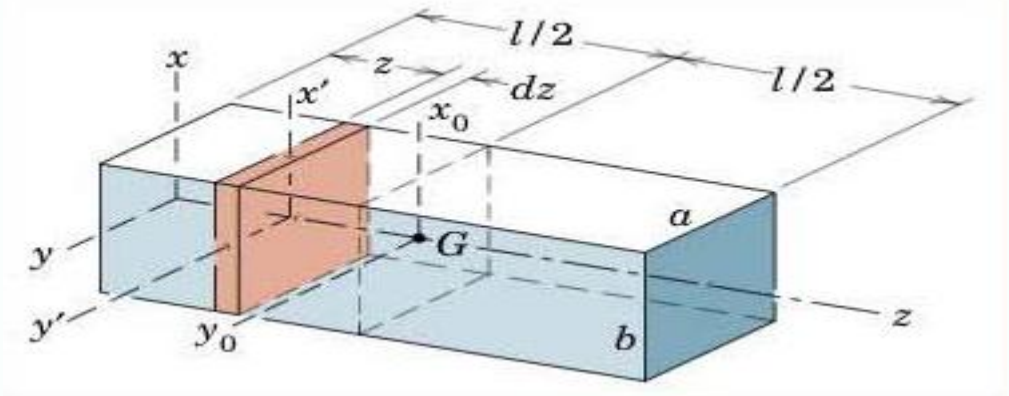
In the figure shown, since y-z is a plane of symmetry.

$$\therefore I_{xy} = I_{xz} = 0$$



Example 2- 2

Determine the moments of inertia of the cuboid shown about the centre of mass G.



$$dm = \rho dV = \rho\, dx\, dy\, dz$$

Gives

$$I_{x_o} = \rho \iiint (y^2 + z^2) dx\, dy\, dz$$

$$I_{y_o} = \rho \iiint (x^2 + z^2) dx\, dy\, dz$$

$$I_{z_o} = \rho \iiint (x^2 + y^2) dx\, dy\, dz$$

$$I_{x_o} = \rho \int_{-\frac{b}{2}}^{\frac{b}{2}} \int_{-\frac{a}{2}}^{\frac{a}{2}} \int_{-\frac{l}{2}}^{\frac{l}{2}} (y^2 + z^2)\, dx dy dz$$

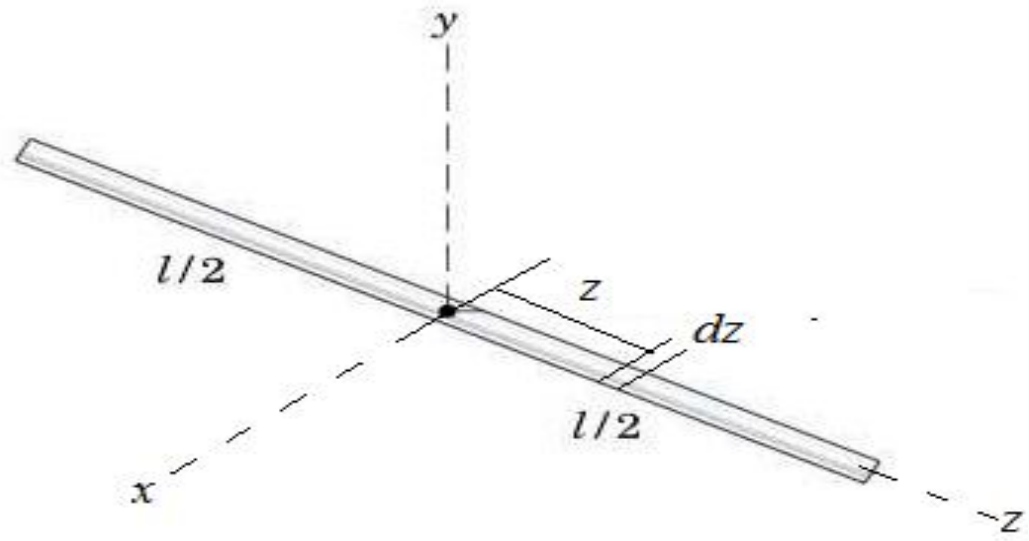
$$I_{x_o} = \frac{m}{12}(a^2 + l^2)$$

$$I_{y_o} = \frac{m}{12}(b^2 + l^2)$$

$$I_{z_o} = \frac{m}{12}(a^2 + b^2)$$

Example 2- 3

Find the moments and products of inertia of a slender of mass m and length l about the x, y and z axes through the centre of mass.



Solution

Consider a uniform slender bar with its length along the y axis as shown. If ρ is its mass per unit length, the mass of the element of length dz is dm = ρdz. Also, if the lateral dimensions are neglected, then

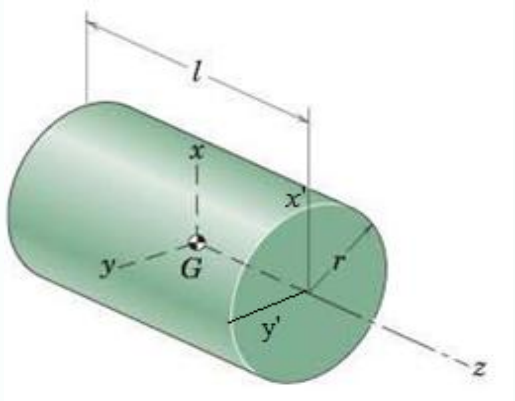
$$I_{zz} = \int_m \left(x^2 + y^2\right) dm = 0$$

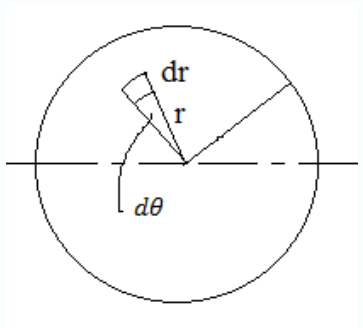
$$I_{xx} = \rho \int_m \left(y^2 + z^2\right) dz = \rho \int_{-l/2}^{l/2} z^2 dz = \frac{ml^2}{12}$$

$$I_{xx} = I_{yy}$$

Example 2- 4

Determine I_{xx}, I_{yy} and I_{zz} of the cylinder shown below





$$dm = \rho dA = \rho\, r dr d\theta$$

$$I_{zz} = \rho \iint (y^2 + x^2) dy dx$$

$$I_{zz} = \rho \int_0^{2\pi} \int_0^{r} r^2 \,(r dr d\theta) = \frac{mr^2}{2}$$

For a thin disc with thickness dz

$$I_{zz} = I_{x'x'} + I_{y'y'};\ but\ \ I_{x'x'} = I_{y'y'}$$

$$I_{zz} = 2I_{y'y'}$$

$$\therefore I_{x'x'} = I_{y'y'} = \frac{mr^2}{4}$$

$$I_{xx} = I_{yy} = \frac{I_{zz}}{2} + \int z^2 dm$$

$$I_{xx} = I_{yy} = \frac{mr^2}{4} + \rho A \int_{-l/2}^{l/2} z^2 dz$$

$$\rho A = \frac{m}{\pi r^2 l} \pi r^2 = \frac{m}{l}$$

$$I_{xx} = I_{yy} = \frac{m}{12}(3r^2 + l^2)$$

Inertia Matrix

The inertia properties of a rigid body are completely defined by the nine parameters, which form a 3 x 3 symmetric matrix. Six of the nine parameters are independent of one another. The inertia matrix, which is also called **inertia tensor**, is defined as

$$[I]=\begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{xy} & I_{yy} & -I_{yz} \\ -I_{xz} & -I_{yz} & I_{zz} \end{bmatrix}$$

Example 2- 5

Find the inertia matrix of Example 2-3

Solution

From Example 2-3,

$$I_{xx}=I_{zz}=\frac{1}{12}ml^2$$

$$[I]=\begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{xy} & I_{yy} & -I_{yz} \\ -I_{xz} & -I_{yz} & I_{zz} \end{bmatrix}=\begin{bmatrix} \frac{1}{12}ml^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{1}{12}ml^2 \end{bmatrix}$$

$$[I]=\frac{1}{12}ml^2\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Transformation of Mass Moments of Inertia- translation of coordinates

PARALLEL-AXIS THEOREM

The parallel-axis may be generalized in matrix form as

$$[I_B] = [I_G] + m[K]$$

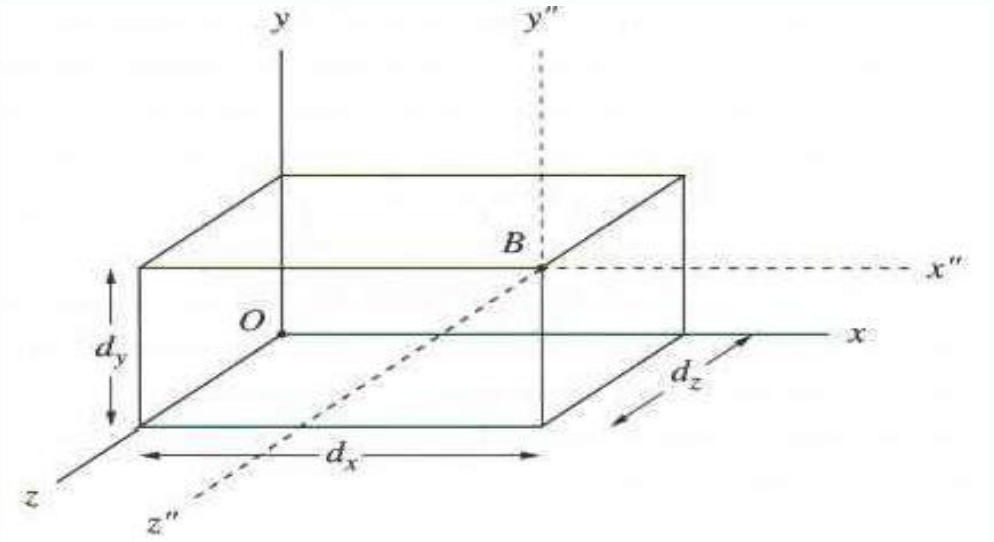
$$[K] = \begin{bmatrix} d_y^2 + d_z^2 & -d_x d_y & -d_x d_z \\ -d_x d_y & d_x^2 + d_z^2 & -d_y d_z \\ -d_x d_z & -d_y d_z & d_x^2 + d_y^2 \end{bmatrix}$$

For Example

$$I_{y"y"} = I_{yy} + m\left(d_x{}^2 + d_z{}^2\right)$$

$$I_{z"z"} = I_{zz} + m\left(d_x{}^2 + d_y{}^2\right)$$

$$I_{x"y"} = I_{xy} + m d_x d_y$$



Radius of Gyration

The radius of gyration of a body is defined as the radius at which the equivalent lumped mass model of the body is located such that the resulting model has the same moment of inertia as the original body. Suppose the radius of gyration of a body of mass m and moment of inertia about axis zz is k_{zz}. From the parallel axis theorem, a concentrated mass m at a radius k_{zz} will have a moment of inertia of

$$I_{ZZ} = m k_{zz}{}^2 \qquad k_{zz} = \sqrt{\frac{I_{ZZ}}{m}}$$

Transformation of Mass Moments of Inertia-rotation of coordinates

Consider the two sets of axes xyz and x'y'z', with the primed axis obtained by rotating the unprimed axis by an angle of □ about the z axis. The two sets of axes are related by

$$x' = x\cos\theta + y\sin\theta$$

$$y' = -x\sin\theta + y\cos\theta$$

$$z = z'$$

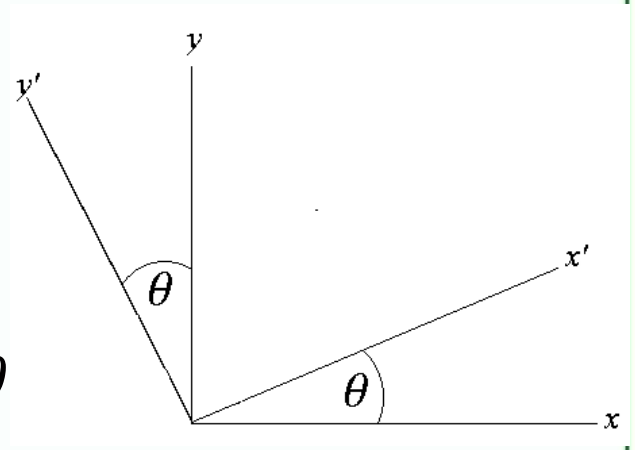
$$\therefore I_{x'x'} = I_{xx}\sin^2\theta + I_{yy}\cos^2\theta - 2I_{xy}\sin\theta\cos\theta$$

$$I_{z'z'} = I_{zz}; \qquad I_{y'y'} = ?$$

$$Similarly, \qquad I_{x'y'} = I_{xy}\cos 2\theta + \frac{I_{xx} - I_{yy}}{2}\sin 2\theta$$

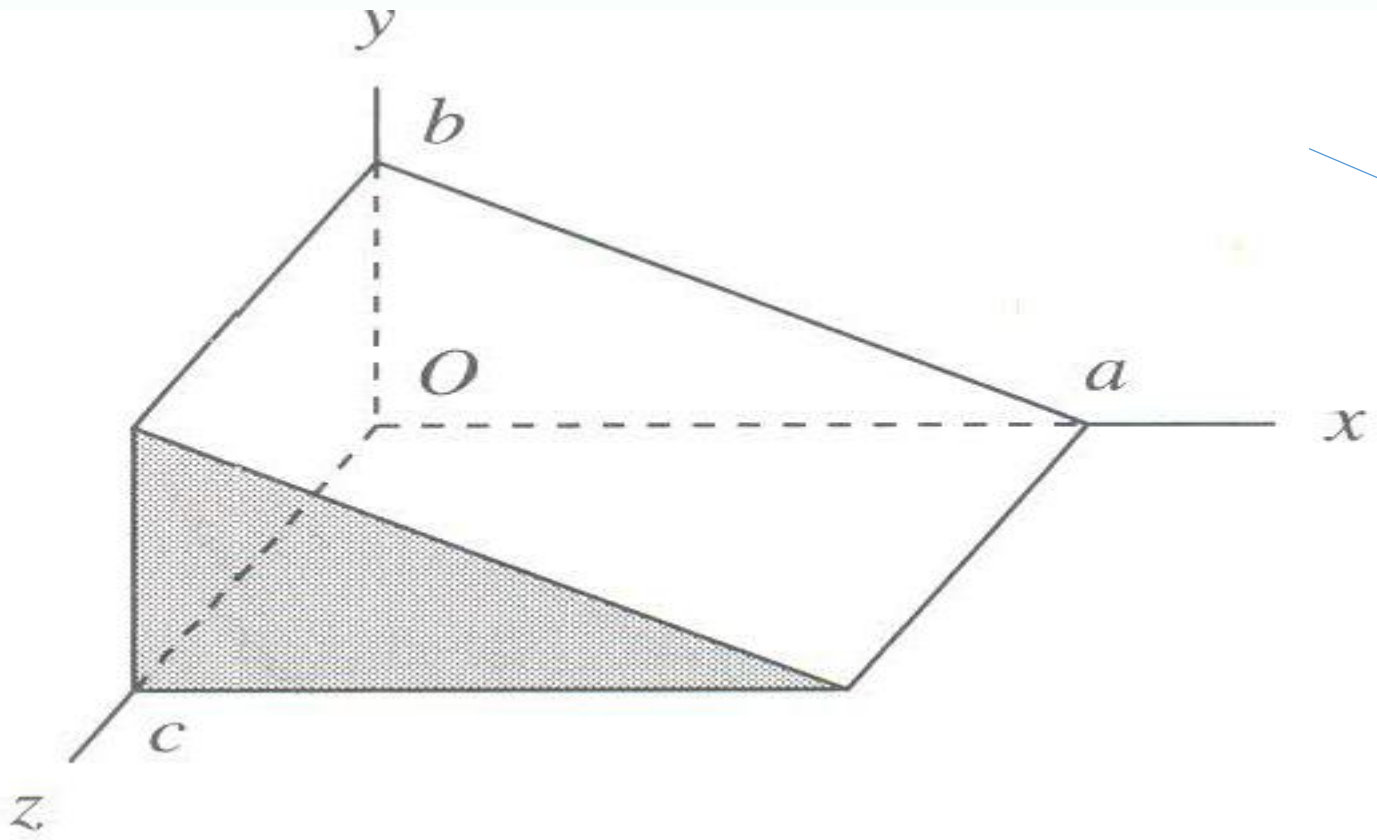
$$I_{y'z'} = -I_{xz}\sin\theta + I_{yz}\cos\theta$$

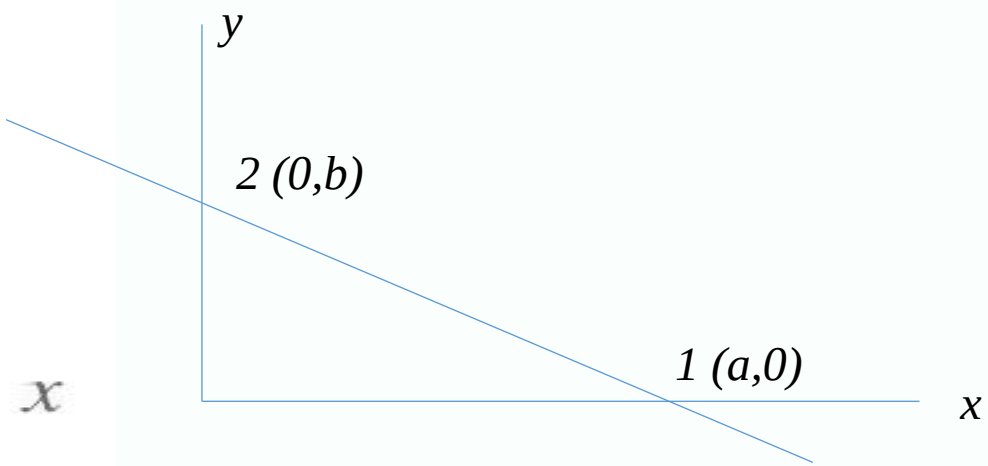
$$I_{x'z'} = I_{xz}\cos\theta + I_{yz}\sin\theta$$



Example 2-6

Find the mass moments of inertia and products of inertia of the right triangular prism of uniform density shown below.





$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y - y_1}{x - x_1}$$

$$\frac{b - 0}{0 - a} = \frac{y - 0}{x - a};\ \ y = \frac{b}{a}(a - x)$$

EXAMPLE 2-5 SOLUTION

To find the moment of inertia, we need to calculate the expressions $\int x^2\,dm$, $\int y^2\,dm$, $\int z^2\,dm$

$$\int x^2\,dm = \rho\int_0^a \int_0^{\frac{b}{a}(a-x)} \int_0^c (x^2)dz\,dy\,dx$$

$$= \frac{1}{a}\rho bc\int_0^a x^2(a-x)dx = \frac{1}{12}\rho a^3 bc = \frac{1}{6}ma^2$$

Similarly, $\int y^2\,dm = \frac{1}{6}mb^2$ $\quad and \quad$ $\int z^2\,dm = \frac{1}{3}mc^2$

$$I_{xx} = \frac{m}{6}(b^2 + 2c^2);$$

$$I_{yy} = \frac{m}{6}(a^2 + 2c^2);$$

$$I_{zz} = \frac{m}{6}(a^2 + b^2);$$

EXAMPLE 2-5 SOLUTION cont'd…

We next find the product of inertias

$$I_{xy} = \rho \int_0^a \int_0^{\frac{b}{a}(a-x)} \int_0^c (xy)\,dz\,dy\,dx$$

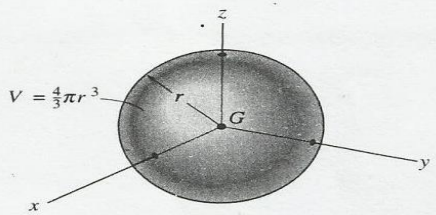
$$= \frac{1}{2a^2}\rho b^2 c \int_0^a x(a-x)^2 dx$$

$$= \frac{1}{2a^2}\rho b^2 c \int_0^a (xa^2 - 2ax^2 + x^3)dx$$

$$= \frac{1}{2a^2}\rho b^2 c \left(\frac{a^2x^2}{2} - \frac{2ax^3}{3} + \frac{x^4}{4}\right)\Big|^a = \frac{1}{12}mab$$

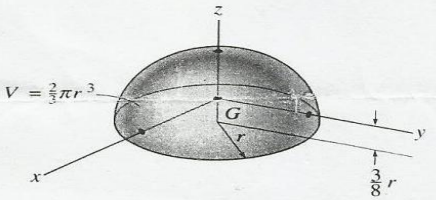
Similarly, $I_{xz} = \frac{1}{6}mac$ and $I_{yz} = \frac{1}{6}mbc$

Center of Gravity and Mass Moment of Inertia of Homogeneous Solids



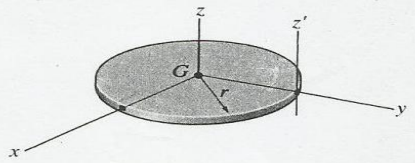
Sphere

$I_{xx} = I_{yy} = I_{zz} = \frac{2}{5} mr^2$



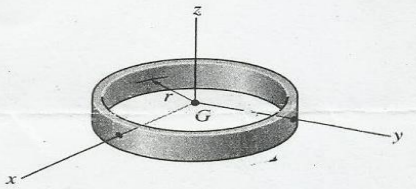
Hemisphere

$I_{xx} = I_{yy} = 0.259mr^2 \quad I_{zz} = \frac{2}{5}mr^2$



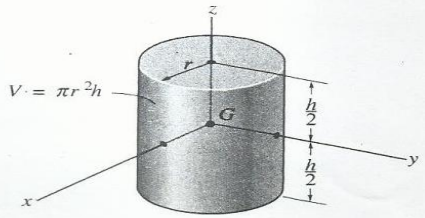
Thin Circular disk

$I_{xx} = I_{yy} = \frac{1}{4}mr^2 \quad I_{zz} = \frac{1}{2}mr^2 \quad I_{z'z'} = \frac{3}{2}mr^2$



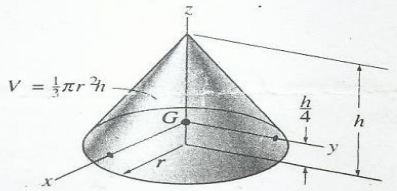
Thin ring

$I_{xx} = I_{yy} = \frac{1}{2}mr^2 \quad I_{zz} = mr^2$



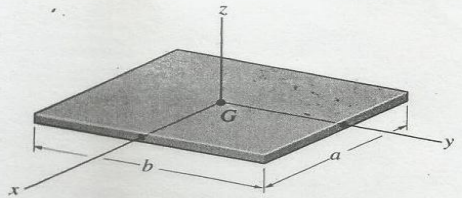
Cylinder

$I_{xx} = I_{yy} = \frac{1}{12} m(3r^2 + h^2) \quad I_{zz} = \frac{1}{2} mr^2$



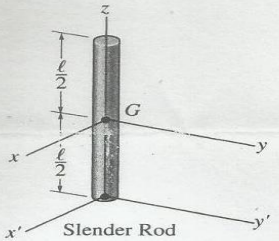
Cone

$I_{xx} = I_{yy} = \frac{3}{80} m\,(4r^2 + h^2) \quad I_{zz} = \frac{3}{10} mr^2$



Thin plate

$I_{xx} = \frac{1}{12}\ mb^2 \quad I_{yy} = \frac{1}{12}\ ma^2 \quad I_{zz} = \frac{1}{12}\ m(a^2 + b^2)$

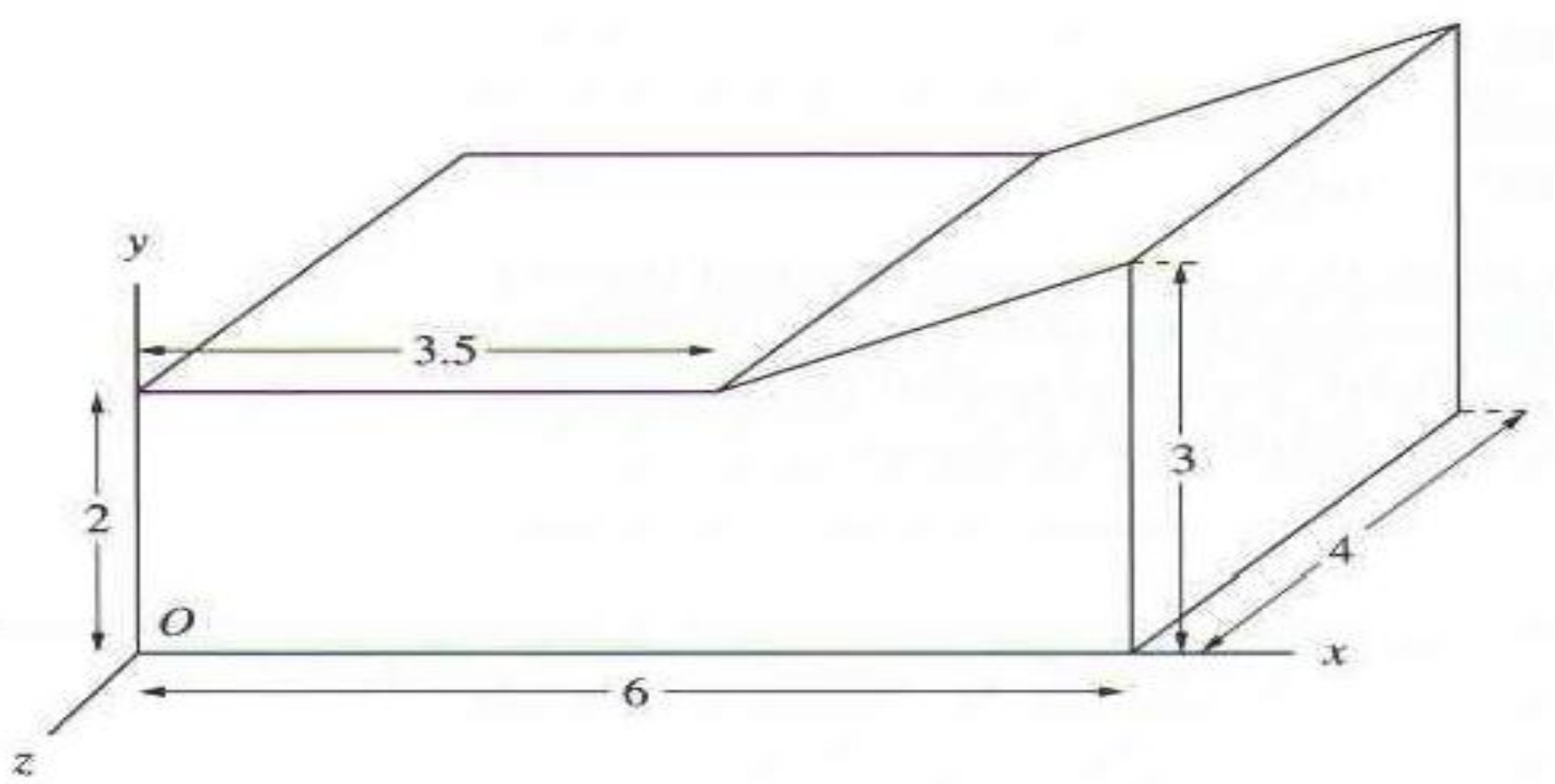


Slender Rod

$I_{xx} = I_{yy} = \frac{1}{12} m\ell^2 \quad I_{x'x'} = I_{y'y'} = \frac{1}{3} m\ell^2 \quad I_{z'z'} = 0$

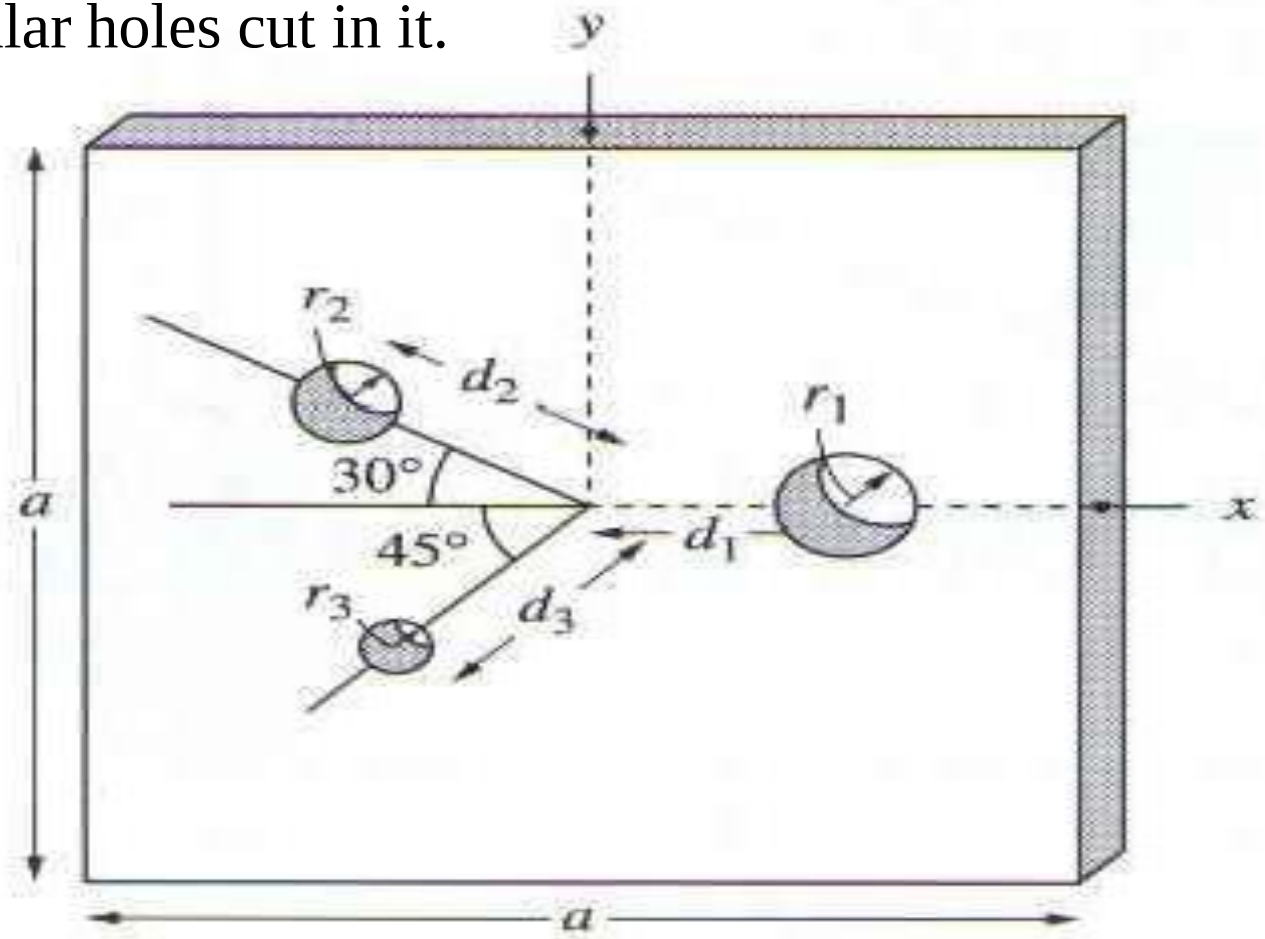
Assignments

1. Find the elements of the inertia matrix associated with the figure Shown about point O, and using the xyz axes.



Assignments cont'd…

2. Determine I_{zz}, I_{xy} and the centroid of the square plate shown with three circular holes cut in it.



a= 100 cm

$r_1 = 10\ cm$

$r_2 = 7\ cm$

$r_3 = 4\ cm$

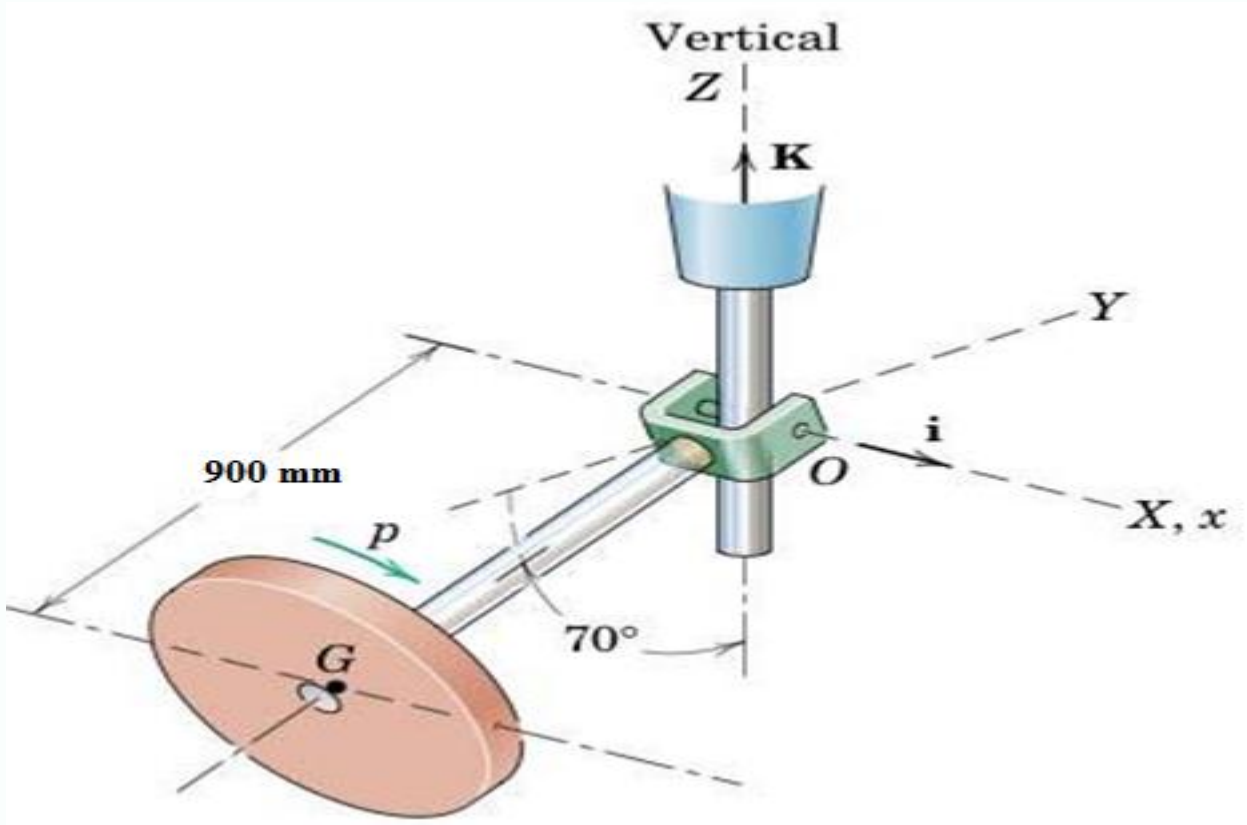
$d_1 = 25\ cm$

$d_2 = 35\ cm$

$d_3 = 30\ cm$

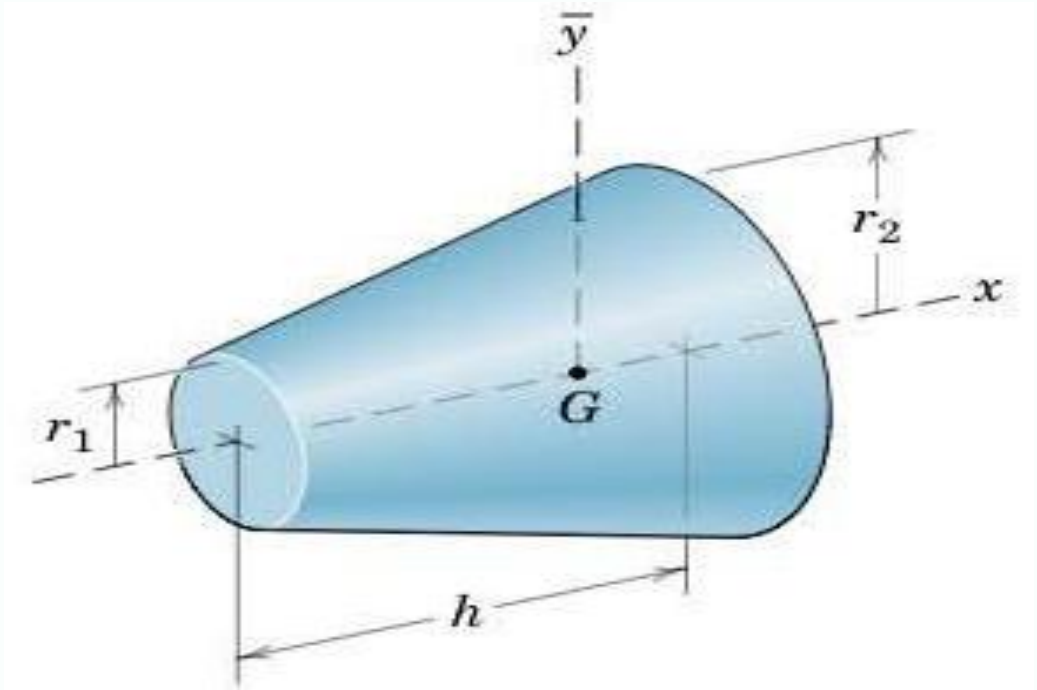
Assignments cont'd…

3. Determine the mass moment of inertia of the disc about the Z axis. Neglect the mass of the shafts. The radius of the disc is 100 mm, the mass is 20 kg and the thickness is 10 mm.



Assignments

4. Determine the mass moment of inertia and centroid of the frustum about the x, y and z axis, the mass of body is 20 kg.



$r_1 = 120\ mm$

$r_2 = 200\ mm$

$h = 300\ mm$

THANK YOU